

Telecom Customer Churn Analysis Using Data Analytics

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Introduction

In the rapidly evolving telecommunications industry, customer retention has become a critical challenge. Customer churn - where customers discontinue their service - directly impacts revenue and growth. Acquiring a new customer costs 5-7 times more than retaining an existing one. This project analyzes telecom customer data to identify key churn drivers and build predictive models to help reduce customer attrition.

Mind Map (Concept Overview)

Customer Data -> Data Cleaning -> EDA -> Statistical Testing -> Feature Engineering -> Predictive Modeling -> Model Evaluation -> Churn Drivers -> Visualize Results -> Generate Insights

Problem Statement

Telecom companies face significant revenue loss due to customer churn, yet many lack a data-driven understanding of why customers leave. There is a need for comprehensive analysis of customer demographics, services, contracts, and billing to identify churn indicators. This project uses statistical testing and classification models to predict churn and uncover actionable insights for reducing attrition.

Project Execution Plan

Week	Task
Week 1	Data collection, understanding, and preprocessing
Week 2	Exploratory Data Analysis (EDA) and statistical testing
Week 3	Feature engineering, model building, and evaluation
Week 4	Power BI dashboard, final report, and PPT presentation

Activity Flow Diagram

Collect Data -> Clean Data -> Analyze Listings -> Detect Patterns -> Identify Risk Indicators -> Visualize Results -> Generate Insights

Milestones

1. Completion of data collection and preprocessing
2. Completion of EDA with visualizations and statistical tests
3. Completion of model building and evaluation (4 models)
4. Creation of Power BI dashboard
5. Preparation and submission of final report and PPT

Tools and Technologies

#	Tool	Purpose
1	Data Sources	CSV / Online telecom datasets (Kaggle)
2	Data Processing	Python (Pandas, NumPy)
3	Analysis	SciPy (Chi-Square, T-test), Scikit-learn, XGBoost
4	Visualization	Matplotlib, Seaborn, Power BI
5	Development	Jupyter Notebook
6	Version Control	GitHub

Cost Management / Estimates

All tools used in this project are free and open-source. Estimated Cost: EUR 0

GitHub Repository

https://github.com/Narmatha-Muralidharan/DA-120B_Telecom_Customer_Churn_Analysis